



Portfolio Optimization with Cumulative Prospect Theory Utility via Convex Optimization

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Abstract

We consider the problem of choosing a portfolio that maximizes the cumulative prospect theory (CPT) utility on an empirical distribution of asset returns. We show that while CPT utility is not a concave function of the portfolio weights, it can be expressed as a difference of two functions. The first term is the composition of a convex function with concave arguments and the second term a composition of a convex function with convex arguments. This structure allows us to derive a global lower bound, or minorant, on the CPT utility, which we can use in a minorization–maximization (MM) algorithm for maximizing CPT utility. We further show that the problem is amenable to a simple convex–concave (CC) procedure which iteratively maximizes a local approximation. Both of these methods can handle small and medium size problems, and complex (but convex) portfolio constraints. We also describe a simpler method that scales to larger problems, but handles only simple portfolio constraints.

Keywords Convex optimization · Cumulative prospect theory · Convex-concave procedure

1 Introduction

1.1 Cumulative Prospect Theory

Analysis of decision-making under uncertainty has long been dominated by von Neumann–Morgenstern (VNM) utility maximization (von Neumann et al., 1944),

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which takes rational behavior as a fundamental assumption. Kahneman and Tversky (1979) observed that the VNM theory fails to explain actual human decision-making behavior in many settings. The subsequently introduced prospect theory (PT) formalizes loss aversion and the overweighting of small probability events, which are inconsistent with VNM utility maximization. To overcome a violation of first-order stochastic dominance in prospect theory, cumulative prospect theory (CPT) was introduced (Tversky & Kahneman, 1992), which replaces probabilities of outcomes with their rank-dependent cumulative probability distribution. This change leads to an overweighting of extreme low-probability outcomes, instead of all low-probability outcomes. Maximizing CPT utility yields more realistic predictions of actual human decision-making behavior than maximizing VNM utility.

1.2 Portfolio Optimization

Our focus is portfolio optimization, i.e., choosing a mix of investments in a set of assets. One approach is based on VNM utility, where the expected value of a concave increasing utility function (which is also concave) is maximized subject to the constraints on the portfolio. Another approach, introduced by Markowitz (1952), poses the problem as a bi-criterion optimization problem, with the goal of trading off the maximization of expected return with minimization of risk, taken to be the variance of the portfolio return. The standard approach is to combine the return and risk, scaled by a risk aversion factor, into a risk-adjusted return, and maximize this concave quadratic objective subject to the constraints. For this reason Markowitz portfolio optimization is also referred to as mean-variance (MV) portfolio optimization. These two approaches are not the same, since the MV utility function is not increasing, but they are closely related. For example, with a Gaussian asset return model and exponential utility, VNM portfolio optimization is the same as MV portfolio optimization (Merton, 1969). In other cases, MV portfolio optimization was shown to be approximately optimal for other forms of utility functions (Levy & Markowitz, 1979).

One advantage of the MV formulation is that the objective can be expressed explicitly as a quadratic function, without an expectation over the asset returns. (The MV objective is the expected value of a function of the return, but one with a simple analytical expression.) This enables it to be solved analytically for special cases (Grinold & Kahn, 1999), and very efficiently using numerical methods for convex optimization when the constraints are convex (Boyd & Vandenberghe, 2004). Leveraging convex optimization, many extensions were developed, such as the inclusion of transaction and holding costs, or multi-period optimization (Boyd et al., 2017),

VNM portfolio optimization generally uses sample based stochastic convex optimization (Shapiro et al., 2021). Here we use samples of the asset returns, which can be historical or generated from a stochastic model of asset returns (presumably fit to historical data). While these methods solve the problem globally, they are substantially slower than MV methods for similar problems.

We now describe these portfolio optimization methods in more detail. We consider a set of n assets. A portfolio is characterized by its asset weights $w = (w_1, \dots, w_n)$, where w_i is the fraction of the total portfolio value (assumed to be positive) invested in asset i , with $w_i < 0$ denoting a short position. The goal in portfolio optimization is to choose w . The general portfolio optimization problem is

$$\begin{aligned} & \text{maximize} && U(w) \\ & \text{subject to} && \mathbf{1}^T w = 1, \quad w \in \mathcal{W}, \end{aligned} \quad (1)$$

with variable $w \in \mathbb{R}^n$. Here $U : \mathbb{R}^n \rightarrow \mathbb{R}$ is a utility function, $\mathcal{W} \subseteq \mathbb{R}^n$ is the set of allowable portfolio weights, and $\mathbf{1}$ is the vector with all entries one. The data in this problem are the utility function U and the portfolio constraint set $\mathcal{W}$, which we assume is convex.

When U is a concave function the portfolio optimization problem (1) is convex, and so readily solved globally (Boyd & Vandenberghe, 2004). When U is not concave, the problem is not convex, and in general difficult to solve globally. In this case, we typically resort to heuristic or local methods, which attempt to solve (1), but cannot guarantee that the globally optimal portfolio is found.

MV portfolio optimization uses the concave quadratic utility function

$$U^{\text{mv}}(w) = w^T \mu - \gamma w^T \Sigma w,$$

the risk-adjusted expected return, where μ is the expected asset return, Σ is the covariance matrix of the asset returns, and $\gamma > 0$ is the risk aversion parameter, used to control the trade-off of the mean and variance of the portfolio return. This yields a convex optimization problem that is efficiently solved.

VNM utility maximization uses a utility function of the form

$$U^{\text{vnm}}(w) = \mathbf{E}u(r^T w),$$

where r is the random asset return vector, and $u : \mathbb{R} \rightarrow \mathbb{R}$ is a concave increasing utility function. Since expectation preserves concavity, U is a concave function of w and this too leads to a convex portfolio optimization problem. In a few cases, the expectation can be worked out analytically, but in most cases one substitutes a sample or empirical average for the expected value, leading to the approximation

$$U^{\text{vnm}}(w) \approx \frac{1}{N} \sum_{i=1}^N u(r_i^T w),$$

where $r_1, \dots, r_N$ are samples of returns. Maximizing this approximation results in a convex portfolio optimization problem.

1.3 This Paper

We consider portfolio optimization under the CPT utility, which we denote U^{cpt} , defined later in Sect. 2. It is well known that CPT utility is not a concave function, so the problem of choosing portfolio weights so as to maximize it is not a convex

optimization problem, as VNM utility maximization and MV portfolio optimization are. This makes it a challenge to carry out CPT utility maximization in practice.

While CPT utility is not concave, we will show that it does have some convexity structure. Specifically, it is the composition of a convex increasing function of concave functions for positive returns, and the composition of a concave increasing function of convex functions for negative returns. This observation allows us to construct a concave lower bound, or minorant, for the CPT utility, and leads immediately to a simple algorithm for maximizing it by repeatedly maximizing the constructed minorant (which is a convex problem, and thus readily solved). This simple minorization–maximization (MM) method leads to a local maximum of the CPT utility (Lipp & Boyd, 2015).

Our MM method scales to medium size problems, with perhaps tens of assets and hundreds of return samples. For larger problems, we give two other algorithms. One algorithm uses a simpler optimization of a minorant to the approximation given by fixing the probability weights that arise in CPT in each step, again relying on iterations that involve solving convex optimization problems. As a result, this method can handle complex portfolio constraints, as long as they are convex. The second additional algorithm scales to very large problems, but handles only simple portfolio constraints. It relies on modern frameworks for automatic differentiation and first-order optimization methods.

Open-source Python implementations of all three methods can be found in the code repository <https://github.com/cvxgrp/cptopt>.

We do not address questions such as whether or when one *should* choose a portfolio that maximizes CPT utility. We only address the question of *how* it can be done, algorithmically and computationally. We provide methods to solve the CPT portfolio optimization problem, but we emphasize that the conceptual framework behind the method is more general. Across computational economics and finance, nonconvex problems frequently arise. The framework for decomposing the problem into convex and concave parts can be extended to such other problems.

1.4 Previous and Related Work

Limited prior work exists on portfolio optimization with CPT utility. Analytical solutions exist for special cases such as single-period settings with one risk-free and one risky asset (Bernard & Ghossoub, 2010; He & Zhou, 2011; Zou & Zagst, 2017) or for two-fund separation under elliptical distributions (Pirvu & Schulze, 2012). Extensions of these special cases to a multi-period setting are considered in Shi et al. (2015). For the general multi-asset cases, heuristics such as particle swarm simulation (Barro et al., 2020), or grid search methods (Hens & Mayer, 2014) are employed, which have been extended to the multi-period case using dynamic programming (De Giorgi & Legg, 2012; Barberis & Huang, 2009). While grid search can accommodate constraints, the particle swarm method used in Barro et al. (2020) cannot, and requires a hyperparameter to be chosen to turn constraints into penalties.

The evaluation of CPT utility along the mean-variance frontier is a commonly used heuristic (Levy & Levy, 2004; Hens & Mayer, 2014). Some authors (e.g.,

Srivastava et al. (2022)) use numerical methods to maximize CPT utility on small problems, do not explicitly mention the numerical solve method, suggesting the use of generic nonlinear optimizers. In contrast, we focus on custom methods that exploit the special structure of the CPT utility maximization problem.

After the initial release of this manuscript, Yan et al. (2022) proposed a method for optimizing a portfolio using CPT utility based on the alternating direction method of multipliers (ADMM) [(see, e.g., Boyd et al. (2011)]. This work is closely related to ours, in that they consider general multi-asset portfolios, and exploit convexity structure, although in a different way than we do. It is the only other method we are aware of that exploits the convexity structure of the CPT utility and can handle constraints. While we make an approximation about the monotonicity of the weights, they employ a method which does not globally solve one of their sub-problems in order to obtain tractable speeds. Carrying out a direct comparison of the methods is not immediately possible since an implementation of the algorithm they describe is not publicly available.

1.5 Outline

We start in Sect. 2 by defining CPT utility, fixing our notation. The CPT utility extends prospect theory (PT) utility, described in Sect. 2.1, by adding a reweighting function, described in Sect. 2.2. In Sect. 2.3 we explore the convexity structure of CPT utility, followed by a description of the CPT utility portfolio optimization problem in Sect. 2.4. In Sect. 3 we describe algorithms that can be used to find a portfolio that maximizes CPT utility. The first method, presented in Sect. 3.1, is a minorization–maximization method that relies on the convexity structure described in the previous section. The second method, described in Sect. 3.2, uses the convex–concave procedure, a method for maximizing the sum of a convex and concave function, and iterates over the probability weights that appear in the CPT utility. The last method, given in Sect. 3.3, is a projected gradient type method, which can scale to large problem sizes. Numerical experiments are presented in Sect. 4, where we evaluate all methods on a toy problem with three assets, a medium-sized problem with more assets, and a large-scale problem, all based on historical asset class data. We give some conclusions in Sect. 5.

2 Cumulative Prospect Theory Utility

2.1 Prospect Theory Utility

In this section we introduce PT utility, the first building block of CPT utility. Like VNM utility, it is monotonically increasing, but PT utility is not concave. PT utility has an inflection point at the origin, which represents a reference wealth. It is concave for positive arguments, i.e., investors are risk averse for gains, and convex on for negative arguments, i.e., investors are risk seeking for losses. Exponential utility

functions are commonly used for both the convex and the concave sections of the PT utility function. We thus define the positive and negative exponential utilities as

$$u_+(x) = 1 - \exp(-\gamma_+x), \quad u_-(x) = -1 + \exp(\gamma_-x),$$

where $\gamma_+, \gamma_- > 0$ are parameters. Here and throughout the paper, functions with a subscript plus sign are applied to gains, and functions with a subscript minus sign are applied to losses. Combining both functions yields the exponential prospect theory utility for a single return

$$u^{\text{prosp}}(x) = \begin{cases} u_+(x) & \text{if } x \geq 0 \\ u_-(x) & \text{otherwise} \end{cases},$$

which is S-shaped. Prospect theory further accounts for loss aversion, which requires $\gamma_- > \gamma_+$, i.e., a marginal decrease in wealth would decrease the utility more than a marginal increase in wealth would increase the utility.

2.2 Probability Reweighting

The second building block of CPT utility is a reweighting function that assigns higher weights to more extreme outcomes. As is common in the CPT literature, we first define the weighting functions $w(p) : [0, 1] \rightarrow [0, 1]$. We take the specific weighting functions

$$w_+(p) = \frac{p^{\delta_+}}{(p^{\delta_+} + (1-p)^{\delta_+})^{1/\delta_+}}, \quad w_-(p) = \frac{p^{\delta_-}}{(p^{\delta_-} + (1-p)^{\delta_-})^{1/\delta_-}},$$

where $\delta_+, \delta_- > 0$ are parameters. We now specify the notion of extreme outcomes. Let $r_1, \dots, r_N \in \mathbb{R}^n$ be the empirical distribution of realized returns on n assets. Consider a vector of portfolio weights $w \in \mathbb{R}^n$, with $\mathbf{1}^T w = 1$, where $\mathbf{1}$ is the vector with all entries one. The associated portfolio returns are $r_1^T w, \dots, r_N^T w \in \mathbb{R}$. Without reweighting, all returns would have equal weight. Let N_- denote the number of negative returns, and N_+ the number of nonnegative returns, with $N_- + N_+ = N$. We let ρ_i denote the returns re-ordered or sorted by the portfolio returns, with index value $i = 1, \dots, N$,

$$w^T \rho_1 \leq \dots \leq w^T \rho_{N_-} < 0 \leq w^T \rho_{N_-+1} \leq \dots \leq w^T \rho_N,$$

i.e., $w^T \rho_1$ is the largest loss and $w^T \rho_N$ is the largest gain. We define the positive and negative decision weights respectively as

$$\pi'_{+j} = \begin{cases} w_+((N_+ - j + 1)/N) - w_+((N_+ - j)/N) & j = 1, \dots, N_+ - 1 \\ w_+(1/N) & j = N_+, \end{cases}$$

$$\pi'_{-j} = \begin{cases} w_-((N_- - j + 1)/N) - w_-((N_- - j)/N) & j = 1, \dots, N_- - 1 \\ w_-(1/N) & j = N_-. \end{cases}$$

We would argue that π'_+ and π'_- should be nondecreasing, i.e., we should put higher weight on more extreme events. This occurs for most reasonable choices of parameters, but there are choices for which monotonicity is (slightly) violated. Thus, we force monotonicity by replacing π'_{+j} with $\min(\pi'_+)$ for all $j < \operatorname{argmin}(\pi'_+)$, and likewise for π'_- . We zero-pad π'_+ and π'_- from the left to be length N , i.e., $\pi_+ = (0_{N_-}, \pi'_+)$ and $\pi_- = (0_{N_+}, \pi'_-)$, where the subscript on the vector zero denotes its dimension. We define for a monotone increasing probability vector π

$$f_\pi(x) = \sum_{i=1}^N \pi_i x_{(i)},$$

which is sometimes called the weighted-ordered-sum or dot-sort function. (The notation $x_{(i)}$ means the i th smallest element of the vector x .) Then, with

$$\phi_+(x) = \max(x, 0), \quad \phi_-(x) = -\min(x, 0),$$

we have the total CPT utility given by

$$U^{\text{cpt}}(w) = f_{\pi_+}(\phi_+(u_+(Rw))) - f_{\pi_-}(\phi_-(u_-(Rw))).$$

2.3 Convexity Properties

In this section we describe some convexity properties of the CPT utility function. PT utility is convex for negative arguments and concave for positive arguments by definition. CPT utility, i.e., with reweighting, is a difference of two structured terms

$$U^{\text{cpt}}(w) = \underbrace{f_{\pi_+}}_{\text{convex}}(\underbrace{\phi_+(u_+(Rw))}_{\text{concave}}) - \underbrace{f_{\pi_-}}_{\text{convex}}(\underbrace{\phi_-(u_-(Rw))}_{\text{convex}}).$$

The first term is a composition of dot-sort-positive, $f_\pi \circ \phi_+$, and the concave exponential utility for gains, u_+ . The dot-sort-positive function is convex, because dot-sort is convex and increasing for positive weights π , and ϕ_+ is convex. The weighted sum in the CPT utility is consistent with dot-sort whenever the weights in the dot-sort function are monotone nondecreasing, i.e., $\pi_1 \leq \pi_2 \leq \dots \leq \pi_N$. Similarly, $f_\pi \circ \phi_-$ is convex following the same reasoning, making $-f_\pi \circ \phi_-$ concave, which is in turn composed with the convex exponential utility for losses, u_- . We note that for each return, only one argument of the difference contributes to the CPT utility, as $\phi_+(x)\phi_-(x) = 0$. These convexity properties motivate principled algorithmic approaches to maximizing the CPT utility, which we explore in Sects. 3.1 and 3.2.

2.4 CPT Utility Portfolio Optimization Problem

The CPT utility portfolio optimization problem is

$$\begin{aligned} & \text{maximize} && U^{\text{cpt}}(w) \\ & \text{subject to} && \mathbf{1}^T w = 1, \quad w \in \mathcal{W}, \end{aligned} \quad (2)$$

with variable w , where $\mathcal{W}$ is the set of feasible portfolio weights. It is not a convex optimization problem, so we will seek approximate solution methods.

We mention some simple methods for solving or approximately solving the CPT utility portfolio optimization problem (2). If the number of assets is very small (say, 3 or 4), we can solve it by brute force computation, by evaluating U^{cpt} over a fine grid of values.

A reasonable heuristic for approximately solving the CPT utility portfolio optimization problem, motivated by Levy and Levy (2004), leverages our ability to efficiently solve the MV portfolio optimization problem. We find the so-called efficient frontier, by solving the MV problem for a number of different values of the risk aversion parameter γ . (This gives the MV efficient frontier.) We evaluate the CPT utility of each of these portfolios, and choose the one with the largest value. While this does not in general solve the problem (2), it often produces a very good, i.e., nearly optimal, portfolio. It can be used as an initial guess for the iterative methods described below. We refer to this method as the MV heuristic for CPT maximization.

3 Optimization Methods

3.1 Minorization–Maximization Method

The CPT utility has the composition form

$$U^{\text{cpt}}(w) = (f_{\pi_+} \circ \phi_+)(u_+(Rw)) - (f_{\pi_-} \circ \phi_-)(u_-(Rw)).$$

We denote a general linearization of a convex (concave) function $h(w)$ at the point $\hat{w}$ as

$$\hat{h}(w, \hat{w}) = h(\hat{w}) + g^T(w - \hat{w}),$$

where g is a subgradient (supergradient) of the function h . As all linearizations that follow occur at $\hat{w}$, we suppress the second argument.

At $\hat{w}$, we create a concave approximation of the first term of $U^{\text{cpt}}(w)$ by linearizing $(f_{\pi_+} \circ \phi_+)$. We approximate the second term in the difference by linearizing the inner convex utility u_- . To linearize dot-sort-positive, we observe that a subgradient is given by the vector g_x with entries

$$g_{x,i} = \begin{cases} 0 & \text{if } x_i < 0 \\ \pi_{+\sigma_x(i)} & \text{otherwise,} \end{cases}$$

where σ_x is the permutation which maps i to the rank of x_i in x . The minorant at $\hat{w}$ is therefore given by

$$\tilde{U}^{\text{cpt}}(w) = (\widehat{f_{\pi_+} \circ \phi_+})(u_+(Rw)) - (f_{\pi_-} \circ \phi_-)(\widehat{u_-}(Rw)).$$

The minorization–maximization (MM) algorithm (also called the majorization–minimization algorithm when solving a minimization problem) simply iterates between creating the minorant at the current iterate and then maximizing it to find the next iterate (Hunter & Lange, 2000). Our minorant is concave, so maximizing it is efficient. Here $\mathcal{W}$ can be any DCP convex constraint set, since each iteration requires solving a general convex optimization problem.

Algorithm 1 Minorization–maximization method

given w^0 , let $\hat{w} := w_0$.

repeat:

1. Let w^{next} be a maximizer of $\tilde{U}^{\text{cpt}}(w)$, subject to $w \in \mathcal{W}$.
2. **break if** $w^{\text{next}} = \hat{w}$.
3. Update $\hat{w} = w^{\text{next}}$.

return $\hat{w}$.

3.2 Iterated Convex–Concave Method

Though the CPT portfolio optimization objective is non-convex, we know the curvature and sign properties of the component functions which are composed to form the utility. In particular, PT utility is convex on the negative reals, and concave on the nonnegative reals. Thus, it is amenable to optimization via the convex–concave procedure (CCP) Lipp and Boyd (2015); Shen et al. (2016); Yuille and Rangarajan (2003); Lanckriet and Sriperumbudur (2009). The convex–concave procedure for maximization iteratively linearizes the second term in the sum of a concave and a convex function, and maximizes this surrogate objective. While the PT utility has this clear curvature, the CPT utility does not, due to reweighting. Our heuristic approach is to fix the probability weights in each iteration, and then solve the fixed weight CPT utility optimization problem with the convex–concave procedure. Once the weights have been fixed, we can write the PT utility as a concave function

$$f^{\text{ccv}}(x) = \begin{cases} 1 - \exp(-\gamma_+ x) & \text{if } x \geq 0 \\ \gamma_- x & \text{otherwise,} \end{cases}$$

plus a convex function,

$$f^{\text{cvx}}(x) = \inf_{z \leq 0, z \leq x} (-1 + \exp(\gamma_- z) - \gamma_- z).$$

Here “cvx” and “ccv” denote convex and concave, respectively. (See §A for a derivation of these functions in disciplined convex programming (DCP) form.) Unlike the MM algorithm in Sect. 3.1 which maximizes a global lower bound, this approximation is only local, so we include a trust region constraint, which we omit from the algorithm description for brevity. Note that as before, $\mathcal{W}$ can be any DCP convex constraint set.

Algorithm 2 Convex–concave procedure

given w_0 , let $\hat{w} := w_0$.

repeat:

1. Let π be the concatenation of the reversed vector π'_- followed by π'_+ , where π'_- and π'_+ are the decision weights associated with $\hat{w}$ (see §2).
2. Let L_i^{cvx} be the linearization of f^{cvx} at $(R\hat{w})_{(i)}$
3. Let w^{next} be a maximizer of $\sum_i \pi_i (f^{\text{ccv}}(w^T \rho_i) - L_i^{\text{cvx}}(w^T \rho_i))$, subject to $w \in \mathcal{W}$, where ρ_i is the row of R associated with $(R\hat{w})_{(i)}$.
4. **break if** $w^{\text{next}} = \hat{w}$.
5. Update $\hat{w} = w^{\text{next}}$.

return $\hat{w}$.

3.3 Projected Gradient Ascent

We first consider maximizing the CPT utility using gradient ascent (GA). While the CPT utility is not differentiable everywhere, we can use an automatic differentiation package such as PyTorch Paszke et al. (2019) to specify the computation chain for the problem and automatically compute the gradient at points where the utility is differentiable, and a reasonable surrogate for the gradient (such as a subgradient for convex functions) at points where it is not. Such libraries are extremely fast and optimized for use on GPUs. We can then perform gradient ascent, together with a method to enforce the portfolio constraints. Projected gradient ascent consists of the iterations

$$w^{k+1} = \Pi(w^k + \eta^k \nabla f(w^k)),$$

where k denotes iteration, $\eta^k > 0$ is a stepsize and Π is ℓ_2 or Euclidean projection onto the constraint set $\mathcal{W}$, i.e., $\Pi(w) = \arg\min_{w' \in \mathcal{W}} \|w' - w\|_2$.

Using these computation frameworks requires the projection to be expressed as a simple computation chain, which can be done in simple cases such as a long-only portfolio, i.e., $\mathcal{W} = \mathbb{R}_+^n$. Another option to handle long-only portfolio constraints is to parametrize nonnegative portfolio weights via a multinomial logistic map,

$$w_i = \frac{\exp x_i}{\sum_j \exp x_j}, \quad i = 1, \dots, n,$$

where x is an unconstrained variable.

4 Numerical Examples

To evaluate the efficiency and performance of the proposed methods, we compare them in a series of numerical experiments with increasing data size. We first compile a data set consisting of $N = 600$ monthly returns, covering the 50-year period from 07–1972 to 06–2022. The $n = 14$ assets consist of equities and fixed income securities from different regions, as well as commodities, as displayed in Table 1. The data was obtained from Global Financial Data (2022). The indices are total return indices, i.e., they include dividends and interest payments. The GFD index methodology extends the index history back in time by combining multiple single indices where necessary. We provide the GFD symbol for each asset class for reference.

4.1 Toy Example

Our first small example uses $n = 3$ assets: US stocks, 10-year US Treasury bonds, and 3-month US Treasury T-bills. We choose the CPT function with parameters

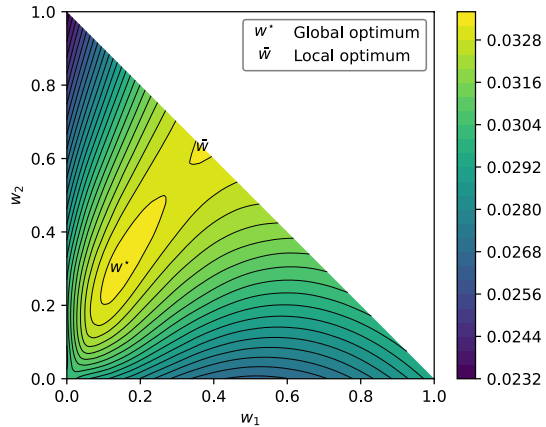
$$\gamma_+ = 8.4, \quad \gamma_- = 11.4, \quad \delta_+ = 0.77, \quad \delta_- = 0.79,$$

which are reasonable, and at the same time exhibit clear non-convexity and even multimodality of the CPT utility. (Many other reasonable choices of the parameters

Table 1 Asset classes and regions in the data set

Asset class	Region	GFD symbol
Equity	US	SPXTRD
Equity	Europe	STOXXER
Equity	Japan	TOPXDVD
Equity	Emerging markets	TRGFDEM
Government bonds	US	TRUSG10M
Corporate bonds	US	TRCCRBD
Government bonds	Europe	TREUROGM
Government bonds	Japan	TRJPNBIM
Bills	US	TRUSABIM
Bills	Europe	TREUROBM
Bills	Japan	TRJPNBIM
Commodities	Global	TRUSACOM
Gold	Global	XAU_BD
Silver	Global	XAG_HD

Fig. 1 CPT utility surface for a long-only portfolios of stocks (w_1), bonds (w_2), and T-bills ($w_3 = 1 - w_1 - w_2$)



lead to unimodal CPT utility, which makes the portfolio optimization problems easy; our goal is to evaluate the methods on more challenging problem instances.)

Figure 1 gives a plot of this utility function for $\mathcal{W} = \mathbb{R}_+^2$, i.e., long-only portfolios. The horizontal axis is w_1 , the fraction invested in stocks; the vertical axis is w_2 , the fraction invested in bonds. The fraction invested in T-bills is $w_3 = 1 - w_1 - w_2$. Thus, the point (0, 0) represents a portfolio fully invested in T-bills. Any portfolio on the diagonal connecting (1, 0) and (0, 1) represents portfolios invested in a convex combination of only stock and bonds. Since there are only two portfolio weights to optimize over, we can find the global maximum using brute-force evaluation of the utility over a fine grid. The global maximum is attained at $w^* = (0.14, 0.3)$, yielding $U^{\text{cpt}}(w^*) = 0.0334$. In addition, there is a local optimum with slightly lower utility at $\bar{w} = (0.37, 0.63)$, which yields $U^{\text{cpt}}(\bar{w}) = 0.0332$.

MV frontier A simple heuristic is to evaluate U^{cpt} on portfolios along the mean-variance efficient MV frontier and choosing the maximizing portfolio among them. Based on the sample mean and covariance of the returns, we first find the

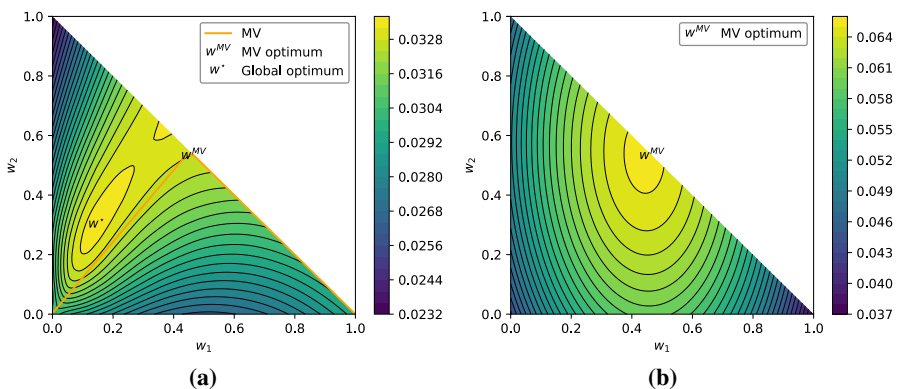


Fig. 2 **a** Maximizing U^{cpt} along the MV frontier, resulting in w^{mv} . **b** Utility surface of U^{mv} for the choice of λ that results in w^{mv}

return-maximizing and risk-minimizing portfolios, and then sample 100 points that are equidistant in volatility space along the efficient MV frontier. Figure 2a shows the efficient MV frontier, and the portfolio with the highest CPT utility along it, w^{mv} , associated with risk aversion parameter $\gamma = 3.2$. It achieves CPT utility of $U^{\text{cpt}}(w^{\text{mv}}) = 0.0328$. Figure 2b shows U^{mv} for the choice $\gamma = 3.2$. It should be noted that the MV frontier is independent of the parameter choice of the CPT utility function, and in general the MV optimum can be far away from a local optimum of CPT.

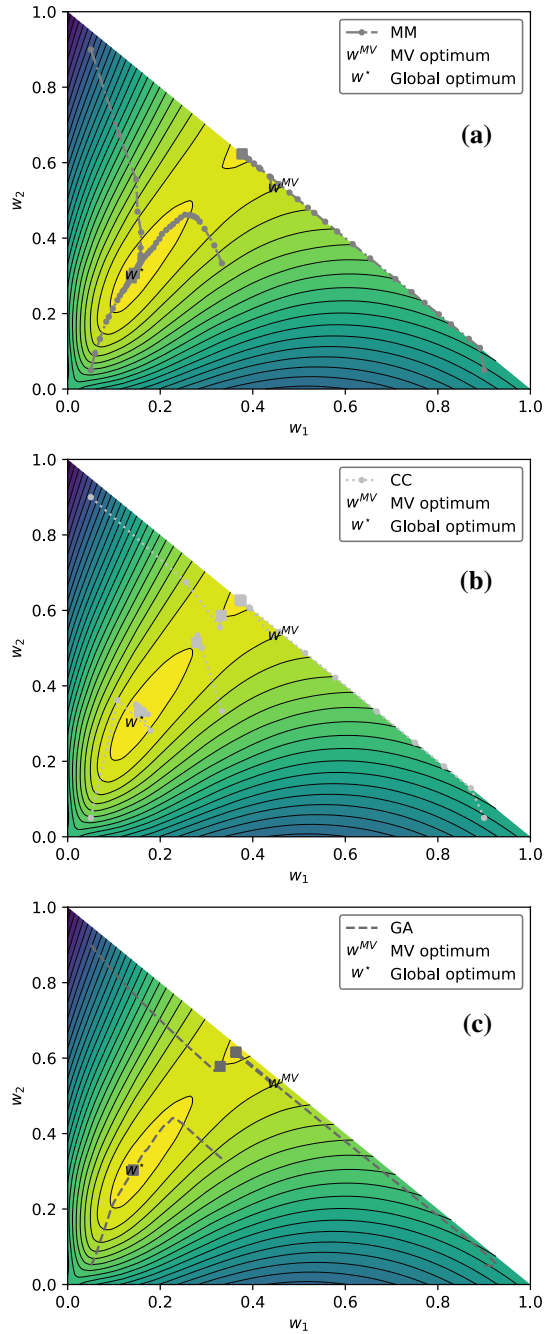
Iterative methods As all remaining methods depend on initialization, we compare the convergence from equal weights, three points close to a full investment in each single asset, as well as the MV optimum in Fig. 3. The MM algorithm terminates at a local maximum from all starting points within fewer than 30 iterations. Likewise, CC converges to a local optimum or a point where the numerical stopping criterion is reached in all cases within at most 11 iterations, albeit on a visually more erratic path. Lastly, the GA method also converges to a point close to a local optimum in all cases. Thus, all iterative methods appear to perform equally well on the toy example.

Diversification To see the effect of the CPT utility on diversification, as well as to understand how the different methods alter the portfolio weights, we run a backtest using a sliding window of 100 observations along our previously described data set of 600 monthly returns. Following (Goetzmann & Kumar, 2008) and (Yan et al., 2022), we compute the sum of squared portfolio weights (SSPW), $\|w - \frac{1}{n}\|_2^2$, as a measure of diversification. We compare the MM, CC, and GA methods to the MV heuristic in Fig. 4. We observe that the MM and CC methods all result in a lower SSPW than the MV heuristic, i.e., the portfolios are more diversified. In addition, we explore how the assumptions of probability reweighting and loss aversion that are inherent to the CPT utility affect the portfolio weights. For this, we change set $\gamma_+ = \gamma_- = 11.4$ in the no loss aversion setting, and $\delta_+ = \delta_- = 1$ in the no probability reweighting setting. We find that both cases result in a much higher median SSPW, indicating that the portfolio weights are less diversified.

4.2 Multi-asset Example

We now extend our example to use all $n = 14$ assets, comparing the achieved utilities, as well as the required computation time. While comparing the absolute wall-times across different implementations can only approximate the computational efficiency of the algorithms, it is relevant to the practicality of the presented methods. The best portfolio on the efficient MV frontier attains a utility of 0.0395 in only 0.6 s. Starting all iterative methods from the equal-weight portfolio, CC terminates first, yielding a utility of 0.0403 in 4.1 s. MM also attains the same utility, but it takes substantially longer, terminating after about 650 s. GA also results in approximately the same utility, being slower than CC, but still dominating MM. When optimizing a single portfolio, we find that GA converges faster using the CPU. However, simultaneous optimization of many portfolios, which is naturally handled by this method, scales better when using a GPU. We use this observation to simultaneously optimize from 10,000 starting points. These weights are sampled from a symmetric Dirichlet distribution

Fig. 3 Convergence from different starting points for the MM (a), CC (b), and GA (c) methods



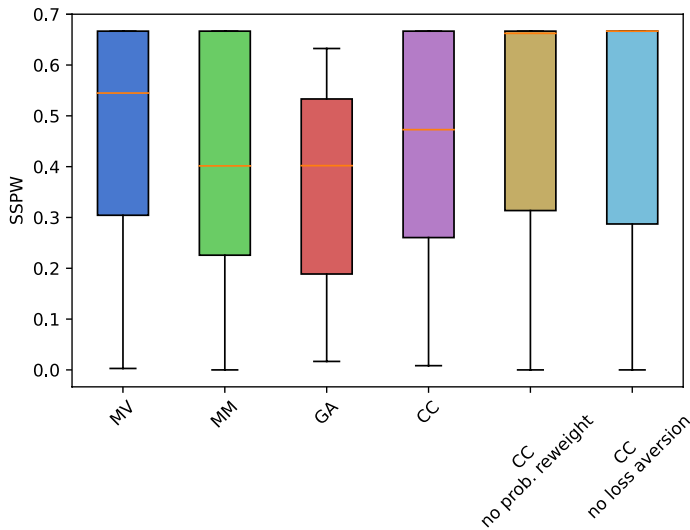


Fig. 4 Comparison of the sum of squared portfolio weights across methods

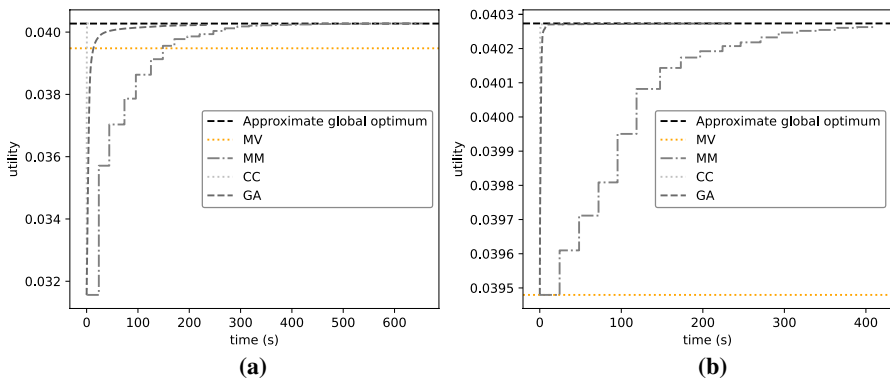


Fig. 5 Comparison of wall-time across methods for the multi-asset example, started from **a** equal-weight portfolio and **b** the best MV portfolio

with concentration parameter $\alpha = 1$, i.e., $w_0 \sim \text{Dir}(\mathbf{1}^n)$, which is equivalent to a uniform distribution over the open standard $(n - 1)$ -simplex. The best resulting utility is denoted as the approximate global optimum, and is not higher than the utilities achieved by all iterative methods when started from equal weights. All methods, as well as the approximate global optimum, are visualized in Fig. 5a. In practice, it would likely be beneficial to leverage the fast computation of the MV portfolio as a starting point for the iterative methods. Indeed, as shown in Fig. 5b, this reduces the time to convergence substantially for all methods, with the GA method now converging in approximately ten seconds. MM is also faster, but still takes about 400 s to converge.

Investigating the sensitivity to the starting point, we sample 30 starting weights and compare the attained utilities. We find that MM and CC converge

to a utility of 0.0403 in all cases. GA, however, displays a higher variance. Its best utility is close to the values obtained by MM and CC. The median utility is 0.0401, which is higher than MV at 0.0395. The worst case utility is 0.0386, which is worse than all other methods.

4.3 Scaling to Many Return Samples

To investigate the scalability of the methods to more return samples, we extend the 600 observations of our data set with synthetic returns. For this, we sample from a Gaussian mixture model with three components that was fit to the return data. We find that GA scales best, handling data sets of hundreds of thousands of observations. For such large data sets, MM and CC did not converge in a reasonable time. Further, as the GA implementation naturally handles optimizing multiple starting points simultaneously, the problem of high variance observed in Sect. 4.2 is alleviated. Figure 6 shows an example where we extend the original data set by a factor of ten, i.e., we consider the case where $N = 6,000$. We choose w^{mv} as a principled starting point for the GA method. We observe that GA does improve over its starting point, however, looking at the axis scale reveals that the improvement over MV is marginal. We observe that the numerical value of the highest utility is different compared to the original data set in Sect. 4.2. This is expected, however, as the observed data only makes up 10% of the extended data set, and the data generating process for the synthetic returns is only an approximation of the true data generating process, which may not be fully described by any single distribution.

5 Conclusions

While the CPT utility is nonconvex and can even be multimodal, we identify some simple convexity properties. Specifically, the CPT utility is a difference of two structured functions, with the first term given by a composition of a convex function with concave arguments and the second term given by a composition of a convex function

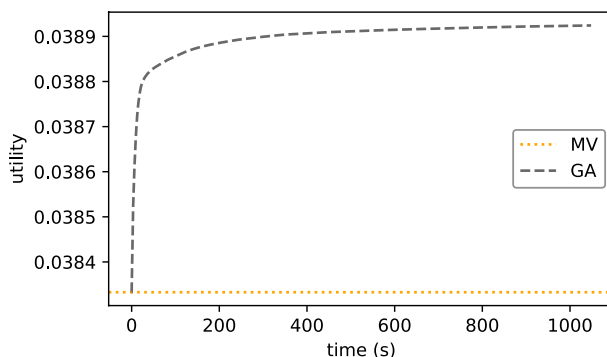


Fig. 6 Convergence of the GA method for $N = 6000$ starting from w^{mv}

with convex arguments. This structure allows us to construct locally tangent concave minorants, which we use to develop a minorization–maximization algorithm to maximize the CPT utility numerically. While the analysis was restricted to the CPT utility, we believe that it motivates similar analyses for other nonconvex objectives commonly used in finance and economics. We provide several practical methods to maximize the CPT utility, including one massively scalable method, and two methods which can easily handle arbitrary convex portfolio constraints. To the best of our knowledge, previous work on maximizing CPT utility considered only simple analytical cases or small problem instances with generic nonlinear optimizers.

As a practical matter, for small problems with arbitrary convex constraints, the MM method has shown smooth convergence and is thus the recommended default method. If this method is too slow, but the portfolio constraints are complex, the CC method should be used instead. For large problems with simple constraints, the GA method appears to be the best choice. As there is low scaling overhead, one should optimize many portfolios simultaneously, including the MV optimal portfolio, an equal weight portfolio, as well as randomly sampled starting points. As all methods are readily available in the accompanying code, it is easy to experiment for the given use case.

Lastly, it is worth noting that the simple method of approximately maximizing the CPT utility by restricting the feasible set to the MV frontier seems to closely approximate the optimal CPT utility in many problem instances.

Appendix

A DCP Form of CCP Objective

A.1 DCP Form of f^{ccv}

To obtain the piecewise definition of f^{ccv} , we split up its argument into a positive and negative part,

$$x = x^+ + x^-, \quad x^+ \geq 0, \quad x^- \leq 0.$$

Now, we observe that

$$f^{\text{ccv}}(x) = 1 - \exp(-\gamma_+ x^+) + \gamma_- x^-$$

is concave, because $\gamma_- > \gamma_+ \geq \frac{\partial(1 - \exp(-\gamma_+ x))}{\partial x}$ for $x \geq 0$. We implement f^{ccv} in DCP form via its hypograph,

$$\begin{aligned} &\{(x, t) \mid f(x) \geq t\} \\ &= \{(x, t) \mid \exists x^+ \geq 0, x^- \leq 0, x = x^+ + x^-, 1 - \exp(-\gamma_+ x^+) + \gamma_- x^- \geq t\}, \end{aligned}$$

which in practice means that to add this function to an optimization problem, we introduce new variables t , x^+ , and x^- , replace f^{ccv} with t , and add the constraints

$$t \leq 1 - \exp(-\gamma_+ x^+) + \gamma_- x^-, \quad x = x^+ + x^-, \quad x^+ \geq 0, \quad x^- \leq 0.$$

A.2 DCP Form of f^{cvx}

To see that f^{cvx} is convex, we can equivalently represent it as a partial minimization of the convex function (of z and x jointly)

$$-1 + \exp(\gamma_- z) - \gamma_- z + \mathbf{1}\{z \leq x\}$$

over the convex set $\{(z, x) \mid z \leq 0\}$. The function can be used in DCP frameworks that provide the indicator function and partial minimization. Alternatively, the indicator function can be omitted when adding the explicit constraints

$$z \leq x, \quad z \leq 0.$$

B Code Snippets

```

1 from scipy.stats import multivariate_normal as normal
2
3 from cptopt.optimizer import *
4 from cptopt.utility import CPTUtility
5
6 # Generate returns
7 corr = np.array([
8     [1, -.2, -.4],
9     [-.2, 1, .5],
10    [-.4, .5, 1]
11 ])
12 sd = np.array([.01, .1, .2])
13 Sigma = np.diag(sd) @ corr @ np.diag(sd)
14
15 np.random.seed(0)
16 r = normal.rvs([.03, .1, .193], Sigma, size=100)
17
18 # Define utility function
19 utility = CPTUtility(
20     gamma_pos=8.4, gamma_neg=11.4,
21     delta_pos=.77, delta_neg=.79
22 )
23
24 initial_weights = np.array([1/3, 1/3, 1/3])
25
26 # Optimize
27 mv = MeanVarianceFrontierOptimizer(utility)
28 mv.optimize(r, verbose=True)
29
30 mm = MinorizationMaximizationOptimizer(utility)
31 mm.optimize(r, initial_weights=initial_weights, verbose=True)
32
33 cc = ConvexConcaveOptimizer(utility)
34 cc.optimize(r, initial_weights=initial_weights, verbose=True)
35
36 ga = GradientOptimizer(utility)
37 ga.optimize(r, initial_weights=initial_weights, verbose=True)

```

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Data availability The code is available at <https://github.com/cvxgrp/cptopt>. The dataset analyzed during the current study are not publicly available due to licensing reasons but are available from the corresponding author on reasonable request.

Declarations

Conflict of interest The authors have no competing interests to declare that are relevant to the content of this article.

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